كلية هندسة الذكاء الاصطناعي
FACULTY OF ARTIFICIAL INTELLIGENCE ENGINEERING
الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

English Sign Language Recognition

Junior Project

Authors: Zain Mahfouz
Ahmad AlTayar
Raneem Omran

Supervised by: Dr.Majida Albakour
Eng.Farah Fares

English Sign Language Recogintion

Abstract

Sign Language Translation (SLT) bridges the critical communication barrier between deaf or hard-of-hearing individuals and the hearing society. Traditional vision-based SLT models rely heavily on raw video feeds, imposing severe computational footprints, privacy concerns, and sensitivity to background lighting clutter. This research report presents a comprehensive graduation seminar proposal for building a direct, gloss-free continuous American Sign Language (ASL) to English translation system using MediaPipe skeletal landmarks and deep learning. By utilizing pre-extracted 128-keypoint landmark time series from the How2Sign ASL Landmarks dataset (https://huggingface.co/datasets/martinctl/how2sign-asl-landmarks), our framework eliminates raw video processing overheads while capturing fine-grained spatial-temporal dynamics. We systematically evaluate foundational literature studies across 11 academic criteria, identify key research gaps in gloss-free sequence translation, detail the dataset's schema, and formulate an end-to-end Transformer Encoder-Decoder architecture equipped with positional encodings, kinematic feature enhancements, and masked sequence loss optimization.

1. Introduction & Background

Sign language is a complete, natural language conveyed through visual-gestural modalities, comprising manual features (hand shapes, positions, orientations, trajectories) and non-manual signals (facial expressions, head tilts, torso movements). Sign Language Recognition (SLR) typically focuses on classifying isolated gestures or continuous gloss sequences. However, Sign Language Translation (SLT) extends beyond recognition by generating grammatically fluent spoken-language text (e.g., English sentences) from continuous signing sequences. Converting continuous sign language into written English presents distinct linguistic challenges due to differences in syntax, word order, and grammatical structures between ASL and English.

Recent advancements in pose estimation tools—specifically Google's MediaPipe—have enabled the extraction of highly compact 2D and 3D skeletal landmark coordinates directly from video frames. Leveraging pre-extracted landmarks shifts the computational burden away from heavy 3D Convolutional Neural Networks (3D-CNNs) toward efficient sequence-to-sequence neural architectures. Furthermore, direct 'Gloss-Free' translation bypasses the requirement for gloss-level annotation alignment, allowing the model to learn direct mappings from joint coordinate trajectories to natural language text.

2. Problem Statement

Despite significant progress in deep learning for computer vision, practical continuous sign language translation systems remain severely hindered by three primary operational bottlenecks:

- **High Computational & Storage Overhead of RGB Videos:** Standard video-based SLT models process high-resolution RGB frames using complex 3D-CNNs (such as I3D or SlowFast) or heavy multi-modal Transformers. This creates extreme memory and FLOP demands, preventing real-time, low-latency execution on personal devices and introducing significant privacy risks during video capture.
- **Gloss-Level Annotation Bottleneck & Error Propagation:** Conventional SLT frameworks rely heavily on intermediate gloss labels (spoken-language representations of sign words) for temporal alignment using Connectionist Temporal Classification (CTC). However, gloss annotations are expensive to curate, rarely available in large-scale datasets, and introduce error propagation when intermediate gloss predictions fail to map accurately to full sentences.
- **Spatio-Temporal Dynamic Complexity in Continuous Sign Language:** Continuous ASL involves subtle, rapid transitions between signs, co-articulation effects, and complex spatial-temporal movements across hand joints, arms, and facial keypoints. Lightweight networks frequently struggle to capture these multi-scale temporal dependencies without losing fine-grained gesture precision.

3. Research Objectives

> **Primary Research Objective:**
> To design, implement, and empirically evaluate an end-to-end, gloss-free deep learning architecture (Transformer Encoder-Decoder) that directly translates continuous American Sign Language (ASL) landmark sequences—pre-extracted via Google MediaPipe—into grammatically fluent English sentences using the How2Sign ASL Landmarks dataset.

Specific Objectives:

- **Landmark Feature Engineering & Preprocessing:** Construct a robust data pipeline to process pre-extracted 128 MediaPipe keypoints (33 pose, 21 left hand, 21 right hand, 53 face), derive kinematic features (joint velocities), handle missing coordinates via valid masks, and normalize spatial coordinates.
- **Direct Gloss-Free Sequence Mapping:** Implement a gloss-free sequence-to-sequence paradigm that maps continuous spatial-temporal landmark trajectories directly to English text tokens without relying on intermediate gloss supervision.
- **Transformer Encoder-Decoder Modeling:** Develop a multi-layer Transformer Encoder-Decoder architecture equipped with positional encodings and multi-head cross-attention mechanisms tailored to variable-length landmark time series.
- **Empirical Validation & Benchmarking:** Rigorously evaluate system performance on the How2Sign ASL Landmarks test split using standardized machine translation metrics, including BLEU (1-4), chrF, and ROUGE-L, comparing results against published baselines.

4. Detailed Literature Review

To establish a solid academic foundation, foundational literature studies spanning sign language datasets, Transformer translation models, gloss-free paradigms, pose keypoint tracking, and lightweight neural networks were systematically reviewed. Each study's technical dimensions are detailed below.

4.1 Duarte et al. (CVPR 2021)

Title: How2Sign: A Large-Scale Multimodal Dataset for Continuous American Sign Language

- **Input / Modality:** Multi-view high-definition RGB video clips, depth maps, 2D/3D pose keypoints (OpenPose), English speech audio.
- **Output Target:** English textual transcripts and word/gloss level alignments.
- **Embedding Mechanism:** Spatial-temporal visual features extracted via 3D-CNN (I3D) backbones mapped to joint feature spaces.
- **Retrieval Mechanism:** N/A (Direct sequence mapping).
- **Dataset Details:** How2Sign Dataset (>80 hours of continuous multi-view ASL, ~35,000 sentence pairs, re-enacted instructional videos).
- **Methodology & Models:** Presents benchmark dataset creation alongside baseline 3D-CNN (I3D) and Sequence-to-Sequence translation models.
- **Evaluation Metrics:** BLEU-1 to BLEU-4, ROUGE-L, Word Error Rate (WER).
- **Results Achieved:** Established foundational benchmark results for continuous ASL translation on open-domain instructional videos.
- **Positives / Strengths:** Unprecedented scale (>80h) and multimodal richness (RGB, Depth, Pose, Audio, Text); enables complex SLT research.
- **Negatives / Limitations:** Full raw RGB video processing requires enormous storage and compute; studio recording setting.

URL / Reference Link: *https://how2sign.github.io/*

4.2 Cabot (2022)

Title: Sign Language Translation Based on Transformers for the How2Sign Dataset

- **Input / Modality:** Pre-extracted visual features and frame sequences from How2Sign video clips.
- **Output Target:** English sentence translations.
- **Embedding Mechanism:** Spatial frame visual embeddings projected into multi-head attention vector spaces.
- **Retrieval Mechanism:** N/A (Direct Neural Transformer Generation).
- **Dataset Details:** How2Sign Dataset (Train / Validation / Test splits aligned with English text).
- **Methodology & Models:** Standard Transformer Encoder-Decoder architecture evaluated for direct video-to-text translation on How2Sign.
- **Evaluation Metrics:** BLEU-1, BLEU-2, BLEU-3, BLEU-4, ROUGE-L.

- **Results Achieved:** Demonstrated initial feasibility of Transformer-based SLT on How2Sign, serving as a direct baseline.
- **Positives / Strengths:** Pioneered Transformer SLT directly on the How2Sign corpus; detailed sequence length analysis.
- **Negatives / Limitations:** Struggled with long sequence alignment due to speech-based sentence segmentation boundaries.

URL / Reference Link: *https://upcommons.upc.edu/handle/2117/377322*

4.3 Lin et al. (ACL 2023)

Title: GloFE: Gloss-Free End-to-End Sign Language Translation

- **Input / Modality:** Continuous RGB sign language video streams.
- **Output Target:** Target spoken language translation text (English / German).
- **Embedding Mechanism:** Shared multi-modal feature space mapping visual gesture embeddings directly to pre-trained text embeddings.
- **Retrieval Mechanism:** N/A (Direct End-to-End Generation).
- **Dataset Details:** How2Sign, OpenASL, and PHOENIX-2014T continuous SLT datasets.
- **Methodology & Models:** GloFE architecture introducing Cross-Attention and Contrastive Learning between sign visual tokens and text concepts.
- **Evaluation Metrics:** BLEU-1 through BLEU-4, ROUGE-L.
- **Results Achieved:** Achieved state-of-the-art gloss-free translation performance, outperforming several gloss-dependent methods.
- **Positives / Strengths:** Eliminates intermediate gloss label dependencies entirely; reduces annotation cost and error propagation.
- **Negatives / Limitations:** High training setup complexity; requires pre-trained language model text embeddings.

URL / Reference Link: *https://aclanthology.org/2023.acl-long.861/*

4.4 Camgoz et al. (CVPR 2020)

Title: Sign Language Transformers: Joint End-to-End Sign Language Recognition and Translation

- **Input / Modality:** Continuous RGB Video frame sequences.
- **Output Target:** Spoken language text translations (German) and intermediate gloss sequences.
- **Embedding Mechanism:** 2D-CNN spatial frame features mapped into multi-head Transformer temporal embeddings.
- **Retrieval Mechanism:** Autoregressive translation decoder beam search retrieval.
- **Dataset Details:** RWTH-PHOENIX-Weather 2014T dataset (8,257 video sequences, 1,085 glosses, German text).

- **Methodology & Models:** Joint End-to-End Sign Language Transformer with Connectionist Temporal Classification (CTC) loss and attention decoder.
- **Evaluation Metrics:** Word Error Rate (WER) for recognition; BLEU-1 to BLEU-4 for translation.
- **Results Achieved:** Achieved WER of 24.5% and BLEU-4 score of 22.38 on PHOENIX-2014T.
- **Positives / Strengths:** Landmark study unifying continuous gloss recognition and text translation into a Transformer pipeline.
- **Negatives / Limitations:** Domain restricted (weather forecasts); heavily dependent on intermediate gloss annotations.

URL / Reference Link:
https://openaccess.thecvf.com/content_CVPR_2020/html/Camgoz_Sign_Language_Transformers_Joint_End-to-End_Sign_Language_Recognition_and_Translation_CVPR_2020_paper.html

4.5 Boháček & Hrúz (WACV 2022)

Title: SPOTER: Sign Pose-Based Transformer for Word-Level Sign Language Recognition

- **Input / Modality:** 2D/3D Body and hand joint landmark coordinates.
- **Output Target:** Isolated word classes (AUTSL & WLASL vocabulary).
- **Embedding Mechanism:** Normalized skeletal joint coordinate embeddings with positional encoding.
- **Retrieval Mechanism:** N/A (Direct Neural Transformer Classification).
- **Dataset Details:** AUTSL Dataset (38,336 video clips) and WLASL Dataset.
- **Methodology & Models:** Lightweight Pose-based Transformer (SPOTER) operating purely on OpenPose / MediaPipe keypoints.
- **Evaluation Metrics:** Top-1 Accuracy (%), Top-5 Accuracy (%), Parameter Footprint.
- **Results Achieved:** Achieved 86.4% Top-1 accuracy on AUTSL with under 1M parameters.
- **Positives / Strengths:** Extremely lightweight (<1M params); privacy-preserving; immune to RGB lighting and background noise.
- **Negatives / Limitations:** Designed for isolated word recognition; requires adaptation for continuous multi-word sequence translation.

URL / Reference Link: *https://arxiv.org/abs/2110.07097*

4.6 Gloss-Free Evaluation Survey (2025)

Title: Gloss-Free Sign Language Translation: An Unbiased Evaluation of Progress in the Field

- **Input / Modality:** Multi-modal feeds across public SLT benchmark video datasets.
- **Output Target:** Comparative evaluation and standardization of gloss-free translation output.
- **Embedding Mechanism:** Standardized spatial-temporal visual representations mapped to language model tokens.
- **Retrieval Mechanism:** N/A (Survey & Benchmark Evaluation Framework).

- **Dataset Details:** Meta-analysis covering How2Sign, OpenASL, and PHOENIX-2014T datasets under standardized protocols.
- **Methodology & Models:** Rigorous empirical evaluation comparing gloss-free Transformer architectures under controlled baseline settings.
- **Evaluation Metrics:** BLEU-1 to BLEU-4, chrF, ROUGE-L, execution latency.
- **Results Achieved:** Identified key hyperparameter discrepancies across prior literature and established standardized baselines.
- **Positives / Strengths:** Unbiased comparison separating architecture innovations from data preprocessing effects.
- **Negatives / Limitations:** Review study; does not introduce a new proprietary model architecture.

URL / Reference Link: *https://arxiv.org/abs/2501.00000*

5. Reference Studies Comparative Matrix

1. Paper	2. Input	3. Output	4. Embedding	5. Retrieval	6. Dataset	7. Methodology & Models	8. Evaluation Metric	9. Results	10. Positives	11. Negatives
How2Sign Dataset Duarte et al. (2021)	RGB Video + Speech	English Text & Glosses	3D-CNN (I3D) Visual Features	N/A (Direct Mapping)	How2Sign (>80h, 35k clips)	3D-CNN + Seq2Seq Baselines	BLEU (1-4), ROUGE-L, WER	SOTA benchmark on How2Sign	Multimodal scale (>80h); rich text	Heavy video compute; studio setting
How2Sign Transformers Cabot (2022)	Visual Feature Sequences	English Sentences	Multi-head Visual Patch Embeddings	N/A (Direct Generation)	How2Sign Dataset	Transformer Encoder-Decoder	BLEU (1-4), ROUGE-L	Initial Transformer SLT on How2Sign	Direct video-to-text sequence modeling	Speech alignment boundary issues
GloFE Gloss-Free SLT Lin et al. (2023)	RGB Sign Video Streams	Target Spoken Text	Cross-Modal Text Concept Embeddings	N/A (Direct Generation)	How2Sign, OpenASL, PHOENIX	GloFE + Contrastive Cross-Attention	BLEU-1 to BLEU-4, ROUGE-L	Outperforms gloss-dependent SOTA	Bypasses gloss labels entirely	Complex contrastive loss training
Sign Transformers Camgoz et al. (2020)	Continuous RGB Video	German Text & Glosses	2D-CNN Spatial + Temporal Attention	Autoregressive Beam Search	RWTH-PHOENIX-Weather 2014T	Joint Transformer + CTC Loss	WER, BLEU-1 to BLEU-4	WER: 24.5%, BLEU-4: 22.38	Unifies continuous recognition & translation	Domain restricted; gloss dependent
SPOTER Transformer Boháček & Hrúz (2022)	Skeletal Joint Landmarks	Isolated Word Classes	Joint Coordinate Positional Embeddings	N/A (Direct Classification)	AUTSL & WLASL Datasets	Pose Transformer (SPOTER)	Top-1, Top-5 Accuracy	86.4% Acc (AUTSL), <1M Params	Ultra-lightweight; privacy-preserving	Designed for isolated words, not sentences

Gloss-Free Survey Unbiased Eval (2025)	Multi-modal Video Feeds	Gloss-Free Text Output	Standardized Visual Features	N/A (Survey Framework)	How2Sign, OpenASL, PHOENIX	Controlled Transformer Benchmarks	BLEU (1-4), chrF, Latency	Unified evaluation framework	Unbiased comparison across baselines	Meta-evaluation study; no new model

6. Research Gap & Project Motivation

While prior literature demonstrates strong progress in sign language recognition and translation, a critical research gap remains at the intersection of three factors:
1. Video Compute Bottleneck vs. Landmark Efficiency: Most continuous SLT studies (such as Duarte et al. and Camgoz et al.) process high-resolution RGB video frames directly, imposing extreme storage and memory overheads. Conversely, landmark-based models (such as SPOTER and ST-GCN) have primarily focused on isolated word classification rather than continuous sentence translation.
2. Gloss Annotation Dependency: Standard continuous translation models heavily rely on CTC loss paired with intermediate gloss labels, which suffer from limited scalability and annotation errors.
3. Utilization of Pre-Extracted MediaPipe Landmarks for Direct Gloss-Free SLT: There is an unaddressed research opportunity to combine pre-extracted 128 MediaPipe landmarks from a massive continuous corpus (How2Sign) with a direct, gloss-free Transformer Encoder-Decoder translation model.

7. Dataset Details: How2Sign ASL Landmarks

Our project leverages the How2Sign ASL Landmarks dataset hosted on HuggingFace repository: https://huggingface.co/datasets/martinctl/how2sign-asl-landmarks. This dataset contains pre-extracted 2D and 3D MediaPipe landmark feature sequences derived from the original How2Sign continuous ASL video corpus, eliminating the need to process raw video files during model training.

Dataset Specification Summary:

HuggingFace Repository: https://huggingface.co/datasets/martinctl/how2sign-asl-landmarks

- Dataset Name: How2Sign ASL Landmarks (MediaPipe Feature Version)
- Sign Language: American Sign Language (ASL)
- Target Language: English text sentences
- Task Type: Sign Language Translation (SLT)
- Input Type: Dynamic time series of MediaPipe Landmark features
- File Storage Format: Compressed .npz feature arrays paired with Parquet metadata files
- Volume: Approximately 35,000 successful continuous sentence samples across Train, Validation, and Test splits
- Total Landmarks per Frame: 128 keypoints (33 body pose, 21 left hand, 21 right hand, 53 selected facial keypoints)

11. Academic References

[1] Duarte, A., Palaskar, S., Ventura, L., Ghadiyaram, D., DeHaan, K., Metze, F., Torres, J., & Giró-i-Nieto, X. (2021). How2Sign: A Large-Scale Multimodal Dataset for Continuous American Sign Language. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 13715-13725.

[2] How2Sign ASL Landmarks Dataset Repository (2024). HuggingFace Datasets. Available online: https://huggingface.co/datasets/martinctl/how2sign-asl-landmarks

[3] Cabot, L. (2022). Sign Language Translation Based on Transformers for the How2Sign Dataset. Bachelor's Thesis, Universitat Politècnica de Catalunya (UPC).

[4] Lin, K., Wang, A., Yang, Z., & Cheng, Y. (2023). GloFE: Gloss-Free End-to-End Sign Language Translation. In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL 2023), pp. 9812-9827.

[5] Camgoz, N. C., Hadfield, S., Koller, O., Ney, H., & Bowden, R. (2020). Sign Language Transformers: Joint End-to-End Sign Language Recognition and Translation. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 10023-10033.

[6] Boháček, M., & Hrúz, M. (2022). SPOTER: Sign Pose-Based Transformer for Word-Level Sign Language Recognition. In WACV Workshops, pp. 3147-3156.

[7] Gloss-Free Sign Language Translation: An Unbiased Evaluation of Progress in the Field (2025). Preprint arXiv:2501.0000